Testicular effects of sub-chronic administration of Hibiscus sabdariffa calyx aqueous extract in rats.

The sub-chronic effect of Hibiscus sabdariffa (HS) calyx aqueous extract on the rat testes was investigated with a view to evaluate the pharmacological basis for the use of HS calyx extract as an aphrodisiac. Three test groups received different doses of 1.15, 2.30, and 4.60 g/kg based on the LD(50). The extracts were dissolved in the drinking water. The control group was given equivalent volume of water only. The animals were allowed free access to drinking solution during the 12-week period of exposure. At the expiration of the treatment period, animals were sacrificed, testes excised and weighed, and epididymal sperm number recorded. The testes were processed for histological examination. Results did not show any significant ($P>0.05$) change in the absolute and relative testicular weights. There was, however, a significant ($P<0.05$) decrease in the epididymal sperm counts in the 4.6 g/kg group, compared to the control. The 1.15 g/kg dose group showed distortion of tubules and a disruption of normal epithelial organization, while the 2.3 g/kg dose showed hyperplasia of testis with thickening of the basement membrane. The 4.6 g/kg dose group, on the other hand, showed disintegration of sperm cells. The results indicate that aqueous HS calyx extract induces testicular toxicity in rats.